

Carsten Wulff

Analog designer, programmer, educator and group manager

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Work experience

Nordic Semiconductor	2015-now	Group Manager, Wireless Recruited 20+ people Active in Bluetooth Special Interest Group; Core Specification Working Group Analog design nRF53 Tool development (Python/C) Embedded programming (nRF52/nRF53)
	2014-2015	Team Lead Analog design nRF91 Tool development (Perl)
	2011-2014	Senior R & D Engineer Analog design nRF52 Tool development (Perl)
	2008-2011	R & D Engineer Analog design nRF51 Tool development (Perl)
NTNU	2021-	Associate Professor TFE4188 - Advanced Integrated Circuits (2022) TFE4152 - Design of Integrated Circuits (2021)
	2020-2021	Guest Researcher Development of TFE4188 - Advanced Integrated Circuits
	2014-2017	Post.Doc A Compiled 9-bit 20-MS/s 3.5-fJ/conv.step SAR ADC in 28-nm FDSOI for Bluetooth Low Energy Receivers , IEEE Journal of Solid-State Circuits
	2004-2008	Ph.D. student Efficient ADCs for nano-scale CMOS technology
	2001-2003	Teaching assistant Planning and execution of transfer of the CAD system from HP-UX to Linux Development of Next Generation Lab, an online electronics lab. Linux administration
Wulff Consulting	2000-2010	Board member and consultant Consulting on .NET and C#
Interconsult group	1999-2000	Summer intern, part time Programming in Visual Basic, Active Server Pages, and PHP.
Lecture experience	2010-now	Guest lectures at NTNU every year, some of the presentations can be found at http://www.wulff.no/presentations
	2006	TFE4200, Analog integrated circuits, responsible for frequency response, noise and CMOS modeling

Programming

I've had a love of programming since my first real summer internship back in 1999. Over the years I've programmed in Visual BASIC, PHP, Ruby, Perl, C#, SKILL, Ocean, Verilog-A, C++, BASH, AWK, VHDL, SPICE, MATLAB, ASP, Java, C, SystemC, Verilog, and probably a few I've forgotten. These days, it's usually Python, C and C++.

2019-now	Python front-end to ciccreator – Python
2009-now	Rapid layout generation tool for SAR ADCs – C++
2017	Power meter blink to kWh using nRF52 – C/Python
2015	Ocean script support package – Perl
2004	SystemDotNet.Report – C#
2005	SystemDotNet: mixed-signal, discrete time, event driven simulator – C#

2003 [Time Tracker 2](#) – C#
2001 [NextGenLab.Chart](#) – C#

Education 2004-2008 **NTNU, Ph.D.**
1997-2002 **NTNU, Master of Science**

Publications [Carsten Wulff on Google Scholar](#)
[Carsten Wulff on IEEE Xplore](#)

Books 2003 Tor A. Fjeldly, Michael S. Shur "Lab on the Web: Running Real Electronics Experiments via the Internet" Chapter 4: Carsten Wulff, Trond Ytterdal, Thomas Aas Sæthre, Arne Skjevlan, ["NEXT GENERATION LABORATORY - A solution for remote characterization of analog integrated circuits"](#), ISBN: 0-471-41375-5, Wiley, 2003

Journal Papers 2017 **Wulff, C**; Ytterdal, T. [A Compiled 9-bit 20-MS/s 3.5-fJ/conv.step SAR ADC in 28-nm FDSOI for Bluetooth Low Energy Receivers](#), IEEE Journal of Solid-State Circuits, Vol: 52, Issue: 7, July 2017
2010 **Wulff, C**; Ytterdal, T. ["Comparator-based switched-capacitor pipelined analog-to-digital converter with comparator preset, and comparator delay compensation"](#), Analog Integrated Circuits and Signal Processing, December 2010 , © Springer
2009 L. Lewyn, T. Ytterdal, C. Wulff, and K. Martin, ["Analog Circuit Design in Nanoscale CMOS Technologies"](#), Proceedings of the IEEE , Vol: 97 , No: 10, 2009 DOI: 10.1109/JPROC.2009.2024663 © IEEE
2009 **Wulff, C**; Ytterdal, T. ["Resonators in Open-Loop Sigma-Delta Modulators"](#), Circuits and Systems I: Regular Papers, IEEE Transactions on, Vol. 56, No. 10, 2009, DOI: 10.1109/TCSI.2009.2015211 © IEEE
2008 **Wulff, C**; Knauserud, Ø; Ytterdal, T. ["Switched Capacitor Analog Modulo Integrator For Application In Open Loop Sigma-Delta Modulators"](#), Analog Integrated Circuits and Signal Processing, February 2008 , © Springer

Conference Papers 2019 Garvik,H; **Wulff, C**; Ytterdal,T [A 68 dB SNDR Compiled Noise-Shaping SAR ADC With On-Chip CDAC Calibration](#) ,IEEE Asian Solid-State Circuits Conference (A-SSCC), 193-194
2017 Garvik,H; **Wulff, C**; Ytterdal,T [An 11.0 bit ENOB, 9.8 fJ/conv.-step noise-shaping SAR ADC calibrated by least squares estimation](#) IEEE Custom Integrated Circuits Conference (CICC), 1-4
2016 **Wulff, C**; Ytterdal, T. ["A compiled 3.5fJ/conv.step 9b 20MS/s SAR ADC for wireless applications in 28nm FDSOI"](#), 42nd European Solid-State Circuits Conference, 2016
2016 Garvik,H; **Wulff, C**; Ytterdal,T [Noise transfer functions and loop filters especially suited for noise-shaping SAR ADCs](#) , IEEE International Symposium on Circuits and Systems (ISCAS), 1034-1037
2009 **Wulff, C.**; Ytterdal, T. ["CBSC pipelined ADC with comparator preset, and comparator delay compensation"](#). Proc. NORCHIP 2009 © IEEE.
2008 Ytterdal, T; **Wulff, C** "On the Energy Efficiency of Analog Circuits in Nanoscale CMOS Technologies", International Conference on Microelectronics (ICM2008). Invited paper.© IEEE.
2008 **Wulff, C.**; Ytterdal, T. ["Design and behavioral simulation of comparator-based switched-capacitor circuits"](#). Proc NORCHIP 2008.© IEEE. [Visual supplements](#)
2007 **Wulff, C.**; Ytterdal, T. ["Design Of A 7-bit, 200MS/s, 2mW Pipelined ADC With Switched Open-Loop Amplifiers In A 65nm CMOS Technology"](#). Proc NORCHIP 2007. ISBN 1-4244-1517-9 © IEEE. [Visual supplements](#)
2006 **Wulff, C.**;Knauserud,Ø; Ytterdal, T. ["Analog Modulo Integrator For Use In Open-Loop Sigma-Delta Modulators"](#). Proceedings of NORCHIP 2006. © IEEE. [Visual supplements](#)
2005 Lillebrekke, C. ; **Wulff, C.**; Ytterdal, T. ["Bootstrapped Switch In Low-Voltage Digital 90nm CMOS Technology"](#). Proceedings of NORCHIP 2005, © IEEE
2005 Berntsen, Ø. ; **Wulff, C.**; Ytterdal, T. ["High Speed, High Gain OTA in a Digital 90nm CMOS Technology"](#). Proceedings of NORCHIP 2005, © IEEE
2005 **Wulff, C.**; Ytterdal, T. ["0.8V 1GHz Dynamic Comparator in Digital 90nm CMOS Technology"](#). Proceedings of NORCHIP 2005, © IEEE
2005 Francisco Colodro, Marta Laguna, Carsten Wulff, Trond Ytterdal and Antonio Torralba. ["New approach for Continuous Time Sigma Delta Modulators"](#). Proceedings of NORCHIP 2005
2004 **Wulff, C.** Erstad, R. Ytterdal, T. ["Programmable Analog Integrated Circuit"](#). Proceedings of NORCHIP 2004. © IEEE
2002 **Wulff, C.** Ytterdal, T. ["Programmable analog integrated circuit for use in remotely operated laboratories"](#). International Conference on Engineering Education (ICEE-2002). [Visual supplements](#)
2002 **Wulff, C.** Ytterdal, T. Sæthre, T.A. Skjevlan, A. Fjeldy, T.A. et al ["Next Generation Lab - A solution for remote characterization of analog integrated circuits"](#). International Caracas Conference on Devices, Circuits and Systems (ICDCS-2002). © IEEE. [Visual supplements](#)

Educational Work	2008	Wulff, C ; Ph.D. Thesis "Efficient ADCs for nano-scale CMOS technology" © Carsten Wulff Pipelined Schematics
	2002	Wulff, C ; Master Thesis "Programmable Analog Integrated Circuit w/Table of Content" © Carsten Wulff
Unpublished Works	2014	Wulff, C ; "The Universal Design Project" , 2014
	2008	Wulff, C ; "An ADC primer" , stuff about ADCs that did not make it into my Ph.D. thesis. This is a rough cut, so there may be some trains of thought that might be difficult to follow.
	2006	Wulff, C ; "FE8113 High Speed Data Converters: A Summary"
	2006	Wulff, C ; "Introduction to Mathematics of Noise Sources"
	2006	Wulff, C ; "On Spectral Density"
	2005	Wulff, C ; "A Brief Introduction to SystemDotNet"
Patents	2018	Reference voltages
	2017	Integrated circuits
	2015	Integrated power-on-reset circuit
	2015	Analogue-to-digital converter
Co-Supervisor		
Ph.D. students	2019-now	Olivia Mirea
	2015-now	Harald Garvik
Master Students	2020	H. Nyholm Localizing Sheep using a Bluetooth Low Energy enabled Unmanned Aerial Vehicle for Round-trip Time of Arrival-based Multilateration
		C. Nielsen ; Ultra Low Power Frequency Synthesizer
	2019	S. Balasubramanian ; Behavioral Modelling and Design of Noise Shaping SAR ADC in 22nm FDSOI
	2015	H. Garvik ; An energy efficient noise-shaping SAR ADC in 28 nm FDSOI
		E. Strandvik ; Compensation of Threshold Voltage for Process and Temperature Variations in 28nm UTBB FDSOI
		T. Nornes Ultra-Low Power SAR-ADC in 28nm CMOS Technologies
	2006	Ø. Knauserud Investigation of errors in open-loop sigma-delta modulators utilizing analog modulo integrators
	2005	Ø. Berntsen High Speed, High Gain OTA in a Digital 90 nm CMOS Technology
		C. Lillebrekke Bootstrapped switches in low voltage nanoscale CMOS process
		L. Røysland Utvikling av testoppsett for en Programmerbar Analog IC
		M. Moksnes Design criteria for a 12-bit 2 Msample/s 1.8 V ADC for realization in 0.18µm CMOS technology
	2004	J. Daleng ; ALEXIS - Automated Laboratory Experiment eXercise Interactive Setup
		F. Strand ; NGAS – Next Generation Analog Station
	2003	V. Pedersen ; NGL webgrensesnitt for FPGA med mikrokontroller
Hobbies		Trail-running and Bouldering